RESHU VERMA

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.75
Rani Laxmi Bai School Chinhat Lucknow	Class XII (CBSE)	2012	86.00%
Rani Laxmi Bai School Chinhat Lucknow	Class X (CBSE)	2010	8.80%

VISITING RESEARCHER

Sakura Science Exchange Program The University of Tokyo, Japan

Dec 2022 (3 weeks)

(Advisor: Prof. Manabu Tsukada)

- Explored Intelligent Transportation Systems and autonomous vehicle safety technologies.

- Analyzed vehicle testbed experiments evaluating active collision avoidance systems.

- Observed emergency braking protocols upon obstacle detection and automatic vehicle re-engagement post-clearance.

SCHOLASTIC ACHIEVEMENTS

- **University Gold Medal (M.Tech CSE):** Achieved the **highest CGPA** in the Computer Science and Engineering department during post-graduation at NIT Uttarakhand.

SKILLS

Languages: C, C++, Node.js, Python

Tools: Hyperledger Sawtooth, Hyperledger Fabric, NS3, SUMO, OMNET++, Veins, Wireshark, MQTT, VS code, LaTeX

Technologies: Permissioned and permissionless blockchain architectures, PBFT consensus, PoW consensus, IoT systems, Intelligent Transportation Systems (ITS), V2X communication protocols, Attribute-Based Encryption (ABE), Schnorr signatures

RELEVANT COURSES

Advance Computer Networks, Advance Data Structure and Algorithms, Cryptology, Internet of Things

PROJECTS

PRANET Lab, Ph.D. Project (IIT Hyderabad: 2020 to 2026)

- **Project 3** - Single-Step Pseudonym Verication and CAM Authentication (SPV-CA) for Vehicular Networks (under review in IEEE T-ITS)
- Proposed SPV-CA, a lightweight pseudonym-based authentication framework that simultaneously provides location privacy, ecient message authentication, and inherent resistance against Sybil attacks in VANETs.
- Developed a single-step Schnorr-signature-based verication mechanism that jointly performs pseudonym verication and CAM authentication with low computational overhead.
- Implemented and evaluated the framework using Veins, SUMO, and OpenSSL-based elliptic curve cryptography.
- **Project 2** - Enhancing Security and Achievability of PBFT-based Consensus Development for Connected Vehicles with Location-based Trust (Published in IEEE T-ITS 2025)
- Proposed V-Trust, a novel solution enhancing location-specic trust in PBFT-based blockchain networks for ITS.
- Introduced Location Certicates (LC) for enabling secure, location-based trust among vehicles.
- Designed a fast joining mechanism leveraging LC verication to eliminate unnecessary PBFT node authentication.
- Enabled context-aware endorser selection to improve the relevance and eciency of consensus development.

- Implemented Delayed Endorsement, allowing vehicles to validate trac events post-departure from the event location.
- **Project 1** - Veriable and Robust Monitoring and Alerting System for Road Safety by AI based Consensus Development on Blockchain (Published in IEEE IV Symposium 2023)
- Integrated blockchain and AI to autonomously detect, alert, verify, and archive dangerous trac events through a collective-intelligence framework.
- Utilized a multi-AI PBFT consensus mechanism to cooperatively verify the contextual validity of detected trac events among vehicles.

CSE Lab, Master's Project (NIT Uttarakhand: 2017 to 2019)

- **Project 3** - Designed an attribute-based authenticated group key transfer protocol without bilinear pairing, using Shamir Secret Sharing and elliptic-curve arithmetic. The protocol provides fine-grained attribute-based access control and securely transfers group keys only to participants satisfying the session initiator's access structure. (Published in Wireless Personal Communications, 2020)
- **Project 2** - Designed a a multi-party attribute-based authenticated key agreement protocol without bilinear pairing, using elliptic curve arithmetic and Lagrange interpolation for fine-grained attribute-based authentication and secure group key establishment.(Published in ICIoTCT, 2019)
- **Project 1** - Designed a two-party attribute-based authenticated key agreement protocol without bilinear pairing, using Shamir Secret Sharing, Diffie-Hellman key exchange, and elliptic-curve arithmetic. The protocol authenticates participants based on an attribute-based access policy and establishes a secure session key for eligible participants. (Published in ICIoTCT, 2019)

JOURNAL ARTICLES

1. **Reshu Verma**, and K. Kataoka, "Enhancing Security and Achievability of PBFT-Based Consensus Development for Connected Vehicles With Location-Based Trust," in ***IEEE Transactions on Intelligent Transportation Systems***, Dec 2025, DOI: 10.1109/TITS.2025.3640392.
2. **R. Verma** and K. Kataoka, "A Single-Step Pseudonym Verification and CAM Authentication Scheme for joint Location Privacy and Sybil Resistance in ITS", in ***IEEE Transactions on Intelligent Transportation Systems, 2026 (under review).***
3. Abhimanyu Kumar, **Reshu Verma** "Attribute-Based Authenticated Group Key Transfer Protocol without Pairing" ***Wireless Personal Communications*** Pub Date : 2020-04-25 , DOI: 10.1007/s11277-020-07292-4

CONFERENCE

1. R. Verma, V. V. S and K. Kataoka, "Veriable and Robust Monitoring and Alerting System for Road Safety by AI based Consensus Development on Blockchain," **2023 IEEE Intelligent Vehicles Symposium (IV)**, Anchorage, AK, USA, 2023, pp. 1-8, doi:10.1109/IV55152.2023.10186676.
2. Reshu Verma and Abhimanyu Kumar "A Pairing Free Attribute-Based Authenticated Key Agreement Pro- tocol Using ECC" February 2020 In book: 4th **International Conference on Internet of Things and Connected Technologies (ICIoTCT), 2019**, DOI: 10.1007/978-3-030-39875-0_11
3. Reshu Verma and Abhimanyu Kumar "Design of Attribute Based Authenticated Group Key Agreement Protocol Without Pairing" February 2020 In book: 4th **International Conference on Internet of Things and Connected Technologies (ICIoTCT), 2019**, DOI: 10.1007/978-3-030-39875-0_10
4. **R. Verma** and K. Kataoka, "Verifiable Uncertain Context (VUC): A Conceptual Framework for Trustworthy Information Management in Intelligent Transportation Systems", ***IEEE ANTS* 2026**. (under review)

REVIEWER PROFILE

1. IEEE IAS Transaction
2. IEEE Internet of Things Journal

TEACHING ASSISTANT EXPERIENCE

- Computer Network 2: Jan 2021 - May 2021, Jan 2022 - May 2022
- Computer Network 1: Aug 2021 - Dec 2021, Aug 2023 - Dec 2023, Aug 2024 - Dec 2024
- Introduction to Programming: Aug 2022 - Feb 2023
- Network Wireless System: Jan 2024 - May 2024
- Blockchain Technology: Jan 2025 - May 2025, Jan 2026 - May 2026